

BUYAGA, Russian Federation

WMO: 310110

Lat: 59.6460N

Lon: 127.1220E

Elev: 913

StdP: 14.22

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-59.0	-55.1	-61.7	0.1	-58.3	-58.1	0.2	-55.0	13.8	-6.7	12.3	-9.2	0.1	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.9	84.4	65.4	80.8	63.6	77.0	62.4	69.0	78.6	66.6	77.2	64.3	73.4	3.9	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	98.8	72.7	63.1	89.3	70.4	60.6	81.8	68.4	33.8	78.3	31.9	77.2	30.1	73.4	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.1	9.4	-62.2	89.6	4.2	3.6	-65.2	92.2	-67.7	94.3	-70.0	96.3	-73.1	98.9		
(5)			-62.6	72.3	4.2	4.2	-65.6	75.4	-68.1	77.8	-70.4	80.2	-73.5	83.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	-32.5	-22.2	0.1	23.3	43.2	57.9	62.9	57.4	41.4	19.4	-13.5	-27.8		
	DBStd	35.99	14.75	13.62	13.92	11.30	7.80	6.56	6.33	6.82	7.27	11.80	14.85	16.06		
	HDD50	12728	2558	2022	1548	802	235	12	1	16	269	949	1904	2412		
	HDD65	17350	3023	2442	2013	1252	675	225	116	250	709	1414	2354	2877		
	CDD50	932	0	0	0	0	26	250	402	244	10	0	0	0		
	CDD65	77	0	0	0	0	0	12	52	13	0	0	0	0		
	CDH74	1483	0	0	0	0	24	417	739	302	1	0	0	0		
	CDH80	394	0	0	0	0	1	90	229	74	0	0	0	0		
	WSAvg	2.4	1.5	1.9	2.0	2.9	3.6	2.7	2.9	2.3	3.0	2.9	1.8	1.8		
	PrecAvg	23.5	0.8	0.7	0.8	1.3	2.2	2.8	3.8	3.6	3.0	2.2	1.4	1.0		
(15) Precipitation	PrecMax	29.9	1.7	1.5	2.3	3.0	4.3	5.3	7.4	8.7	7.0	4.1	3.0	2.2		
	PrecMin	17.8	0.1	0.2	0.2	0.5	0.5	0.4	1.2	0.6	0.6	1.1	0.6	0.4		
	PrecStd	3.2	0.4	0.3	0.4	0.6	0.8	1.0	1.7	1.8	1.3	0.7	0.6	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.9	18.7	42.7	58.8	77.6	87.3	90.6	86.2	69.5	49.1	34.7	21.2		
		MCWB	9.1	16.8	34.3	44.7	57.6	65.9	67.8	66.4	56.0	41.3	32.6	19.9		
	2%	DB	3.2	13.3	34.7	51.1	71.3	82.3	86.1	81.9	63.6	42.4	23.6	13.2		
		MCWB	2.4	11.8	28.8	39.9	53.7	62.3	67.3	63.7	52.4	37.7	22.3	12.1		
	5%	DB	-3.2	6.2	30.4	47.4	65.2	79.0	81.6	77.3	59.1	38.2	15.6	3.9		
		MCWB	-3.8	5.2	26.4	37.0	50.4	61.6	66.1	62.6	50.6	34.5	14.4	3.3		
	10%	DB	-9.7	0.7	25.7	42.6	60.0	75.2	77.6	72.5	54.9	34.6	8.5	-4.8		
		MCWB	-10.1	-0.3	22.5	34.4	47.6	59.6	64.2	60.7	48.2	31.7	7.6	-5.4		
	0.4%	WB	9.1	17.0	34.6	45.0	58.4	71.1	73.4	69.8	57.6	43.1	32.2	19.9		
		MCDB	9.9	18.5	42.2	56.5	75.4	78.8	81.5	79.5	65.7	46.9	34.8	21.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	2.6	11.9	29.6	40.7	54.6	66.2	69.8	66.7	54.5	38.7	22.8	12.3		
		MCDB	3.4	13.4	33.8	50.0	67.5	77.3	78.6	76.9	61.8	42.0	23.7	13.3		
	5%	WB	-3.7	5.3	26.3	37.5	51.8	63.5	68.0	64.4	51.7	35.0	14.4	3.4		
		MCDB	-3.1	6.2	30.3	45.8	63.2	74.7	78.3	73.9	57.1	37.9	15.5	4.0		
	10%	WB	-10.1	-0.1	22.4	34.9	49.1	61.0	65.9	61.9	48.9	31.8	7.7	-5.2		
		MCDB	-9.7	0.8	25.9	42.6	58.3	71.8	76.3	70.7	54.2	34.4	8.5	-4.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.6	27.5	35.9	30.3	24.2	26.4	23.9	25.1	21.5	17.7	19.1	16.6		
		MCDBR	22.5	28.2	36.1	31.5	35.4	35.4	31.5	34.3	29.7	19.0	22.2	20.7		
		MCWBR	21.8	26.9	30.0	20.1	19.5	17.9	14.8	17.7	19.1	13.9	20.6	20.0		
		MCDBR	22.6	28.3	34.3	29.2	31.2	30.1	27.3	28.0	27.7	17.5	22.0	20.8		
	5% WB	MCWBR	21.9	26.9	28.8	19.8	18.5	17.8	15.5	14.9	18.9	13.5	20.6	20.1		
(40) Clear-Sky Solar Irradiance	taub		0.243	0.276	0.312	0.381	0.453	0.430	0.496	0.445	0.346	0.300	0.247	0.220		
	taud		1.984	2.034	2.031	1.961	2.028	2.231	2.042	2.163	2.413	2.271	2.016	1.921		
	Ebn at Noon		164	225	254	254	245	255	231	235	247	219	159	124		
	Edh at Noon		16	26	37	48	48	40	48	39	26	21	15	11		
(44) All-Sky Solar Radiation	RadAvg		180	516	1067	1630	1621	1818	1641	1288	814	426	233	103		
	RadStd		12	22	38	61	116	144	129	95	65	46	17	4		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page